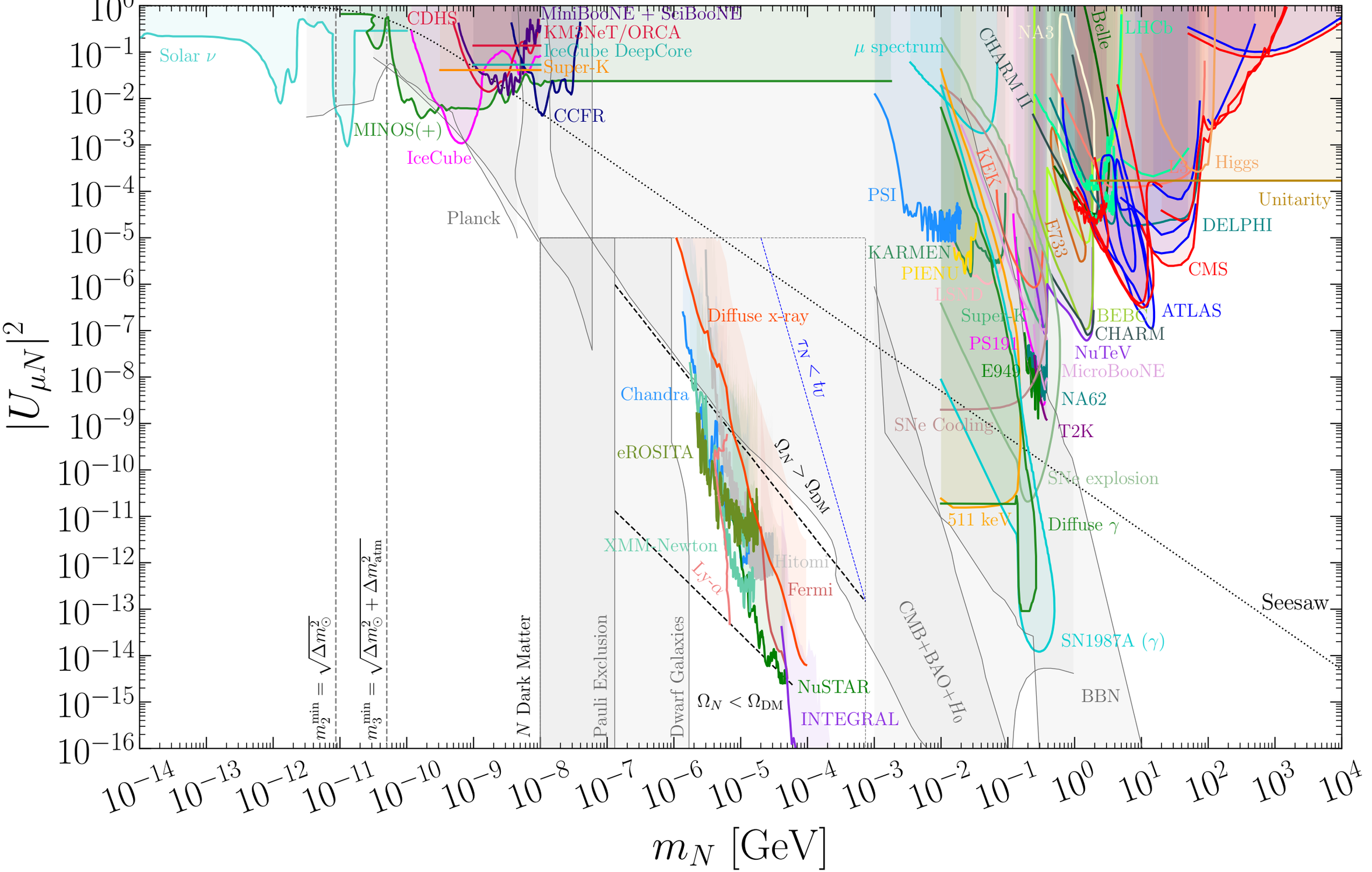

$|U_{\mu N}|^2$
m_N [GeV]
Solar ν
CDHS
MiniBooNE + SciBooNE
KM3NeT/ORCA
IceCube DeepCore
Super-K
CCFR
MINOS(+)
IceCube
Planck
μ spectrum
CHARM II
NA3
Belle
LHCb
Higgs
Unitarity
PSI
KEK
E733
DELPHI
KARMEN
PIENU
LSND
CMS
Super-K
ATLAS
PS191
BEBC
CHARM
NuTeV
E949
MicroBooNE
NA62
T2K
SNe Cooling
SNe explosion
511 keV
Diffuse γ
SN1987A (γ)
BBN
CMB+BAO+H_0
Seesaw
Diffuse x-ray
$\tau_N < t_U$
Chandra
eROSITA
$\Omega_N > \Omega_{\rm DM}$
XMM-Newton
Ly-α
Hitomi
Fermi
NuSTAR
$\Omega_N < \Omega_{\rm DM}$
INTEGRAL
$m_2^{\rm min} = \sqrt{\Delta m^2_\odot}$
$m_3^{\rm min} = \sqrt{\Delta m^2_\odot + \Delta m^2_{\rm atm}}$
N Dark Matter
Pauli Exclusion
Dwarf Galaxies